

Alberto Andrés Valdés González.

Degree: Mathematical Engineer.

Work position: ML-Engineer.

Mail: anvaldes@uc.cl/alberto.valdes.gonzalez.96@gmail.com

Location: Santiago, Chile.

Utterance, Intent, Slot, Keyword, Keyphrase

Intent, Utterance, Slot

Utterance: The actual phrase spoken or typed by the user. There can be many utterances for a single intent.

Examples:

- I want to book a room in Santiago.
- I need a hotel in Santiago.
- Can you find me a place to stay in Santiago?.

Intent: Represents what the user wants to do — their goal or purpose. It's a category of action.

Examples:

- Book hotel.
- Check weather.
- Order food.

Slot: These are the key entities or parameters in the utterance that provide necessary information to fulfill the intent. Used to extract relevant data.

Example: For the intent **book hotel**

- Utterance: I want a hotel in Santiago on June 5th.
- Slots: City = Santiago, Date = June 5th.

Keyword, Keyphrase

Keyword: A single important word that captures the main idea or topic of a text. Used to identify subjects or enable searches.

Example: In a biology article, keywords could be: DNA, enzyme, gene.

Keyphrase: A multi-word expression that conveys more specific or contextual meaning than a single keyword.

Examples: DNA sequencing, gene expression.

Summary:

- Keyword = Hotel, Santiago.
- Keyphrase = Hotel in Santiago, Cheap Hotel Booking.

Concept	Intent-Utterance-Slot	Keyword-Keyphrase
Used in	Conversational assistants / chat-bots	Information retrieval / text analysis
Purpose	Understand user goals	Identify main ideas in a text
Focus	Semantics & conversation structure	Term relevance and frequency
Extracted Elements	Action (intent) + details (slots)	Words or phrases of high informational value

Cuadro 1: Comparison between Intent-Utterance-Slot and Keyword-Keyphrase
